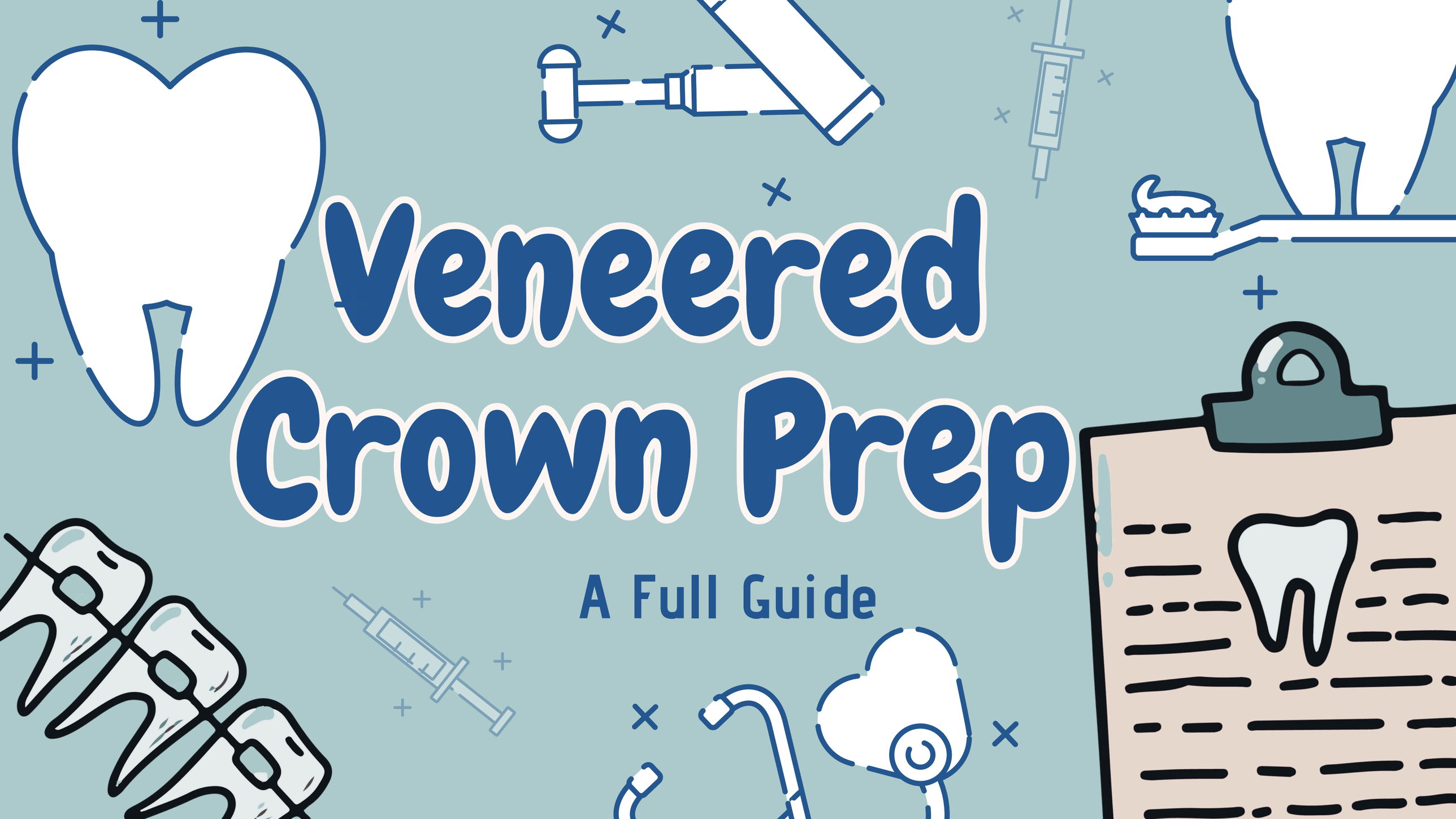

Veneered
Crown Prep
A Full Guide

What is a Veneered Crown Prep?

A Veneered Crown Prep is the type of prep done when the future crown is planned in an **aesthetically important area** .ie: from **second premolar to second premolar**. Or if the patient has esthetic concerns and refuses a full metal crown,.

<u>In this case, the crown is made of two layers</u>

- A base **metal layer** of approx **0.5 mm thickness.**
- A **porcalien/ ceramic layer** only on the buccal surface of approx **1 mm thickness.**

This is where the name veneered comes from. A ceramic veneer is made on the buccal surface while the rest of the crown has the metal layer only.



The picture shows the lingual/palatal surface of a partially veneered bridge.

Preparation Criteria

A veneered crown prep has the **same criteria** as the previously explained full metal crown prep but with a **few extra considerations:**

- Be aware that **all surfaces will be covered with a metal base** while **only the buccal and part of the proximal surfaces will have an additional porcalin layer.**
- This means that the **amount of reduction** and therefore **finish line thickness** will **not be unified across the prep.**
- Veneered preps can have two forms

Both preps are technically correct but you need to verify from your instructor which prep does your school accept.
In the coming videos technique will focus on wingless because it is the most widely used in clinical situations.

winged prep



wingless prep



Measuring the Amount of Reduction

To measure the amount of reduction accurately, use a condensation silicon impression using the technique demonstrated in the video below:



What you need for the perfect prep



Micromotor for low speed



High/low speed Contra



Mirror+ explorer



Probe



Tapered with round end blue



or Tapered with flat end blue



Needle blue



Flame/football blue



Flame/football yellow



Tapered with round yellow

The Occlusal Surface

Occlusal surface Checklist:

- Maintaining anatomy especially the **cusp ridges**
- Reducing the correct amount:

*The **functional cusp needs more reduction** because **more occlusal forces fall on it.**

More reduction=more space for a greater thickness of the material to withstand occlusal forces.

*Functional Cusp= **1.5-2mm**

*Non-Functional Cusp= **1-1.5mm**

- Functional Cusp Bevel
- Smooth with no sharp angles.



The picture shows ideal occlusal reduction where the occlusal anatomy is preserved

The Occlusal Surface

Let's reduce the occlusal surface of an upper premolar together step by step using the tips in the previous slide



The Occlusal Surface



Axial Reduction

Axial Reduction in Veneered prep is the **KEY** step that diffrentiates this type of prep from the full veneer prep.



The main rules are basically the same:

- Identify the Long Axis of the Tooth
- Move your contra **PARALLEL** to the long axis of the tooth **at all times.**
- **NEVER** aim to “draw” the finish line – **it will appear naturally when you reduce the correct amount using the correct hand angulation.**

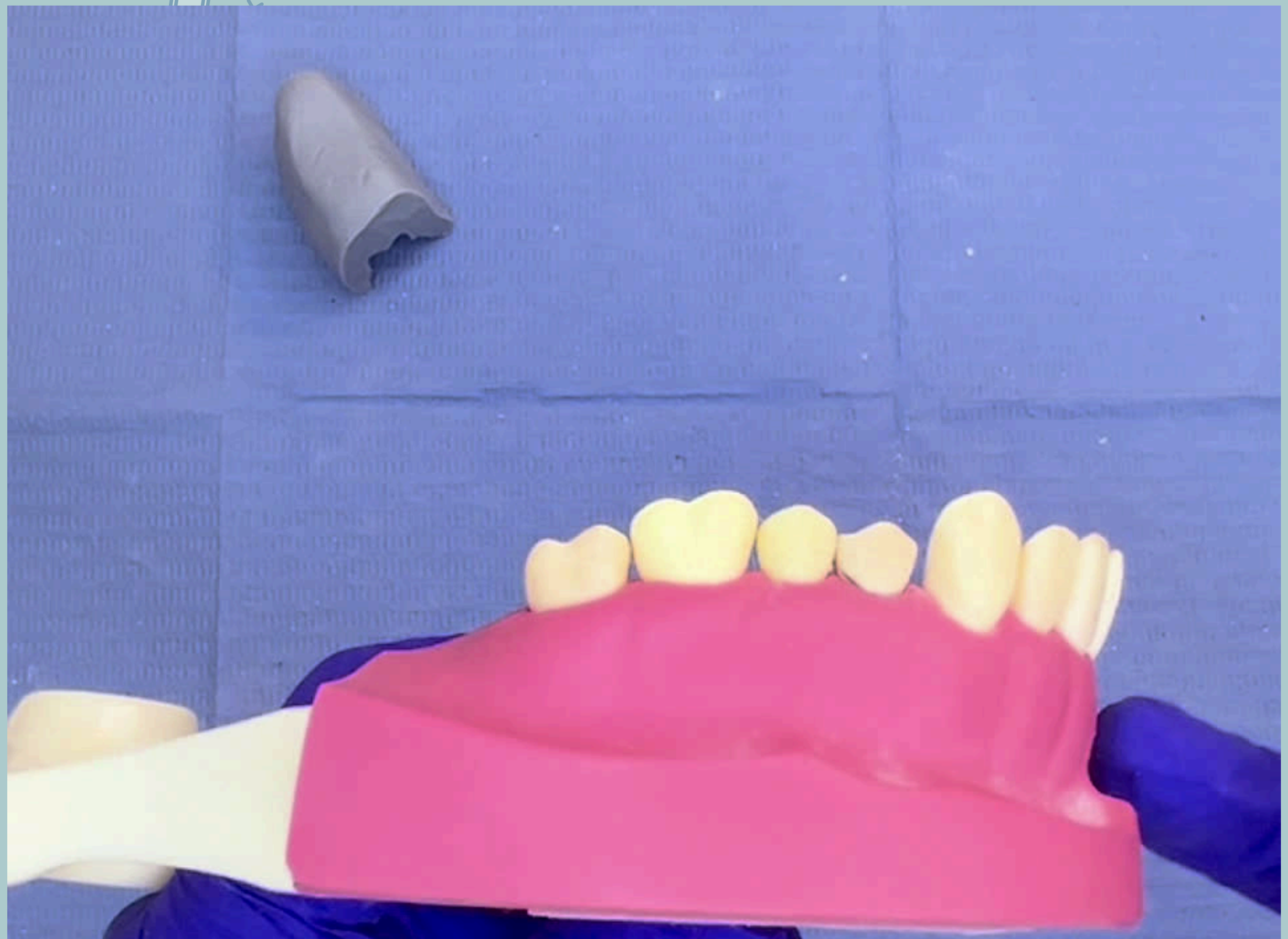
So, what is done differently in a veneered crown prep?

In a veneered crown prep the future crown is made up of 2 different materials remember?
All of the surfaces will get a metal layer. The buccal and part of the proximal surfaces will get an **EXTRA** layer of porcalin/ceramic
so extra preparation is needed on these surfaces to make space for the ceramic layer.

In summary:

- **Start the prep** by reducing all the surfaces by **0.5mm** – this gives space for the **metal layer.**
- Next, go back to the **buccal surface** and reduce it an **additional 1mm** – now the **total finish line thickness buccally is 1.5 mm leaving enough space for both metal and ceramic.**
- The **proximal** surfaces also recieve an **additional 1mm** of reduction BUT only **just before the central groove.** Sounds confusing we know but you will get it better by watching the video on the next slide :)

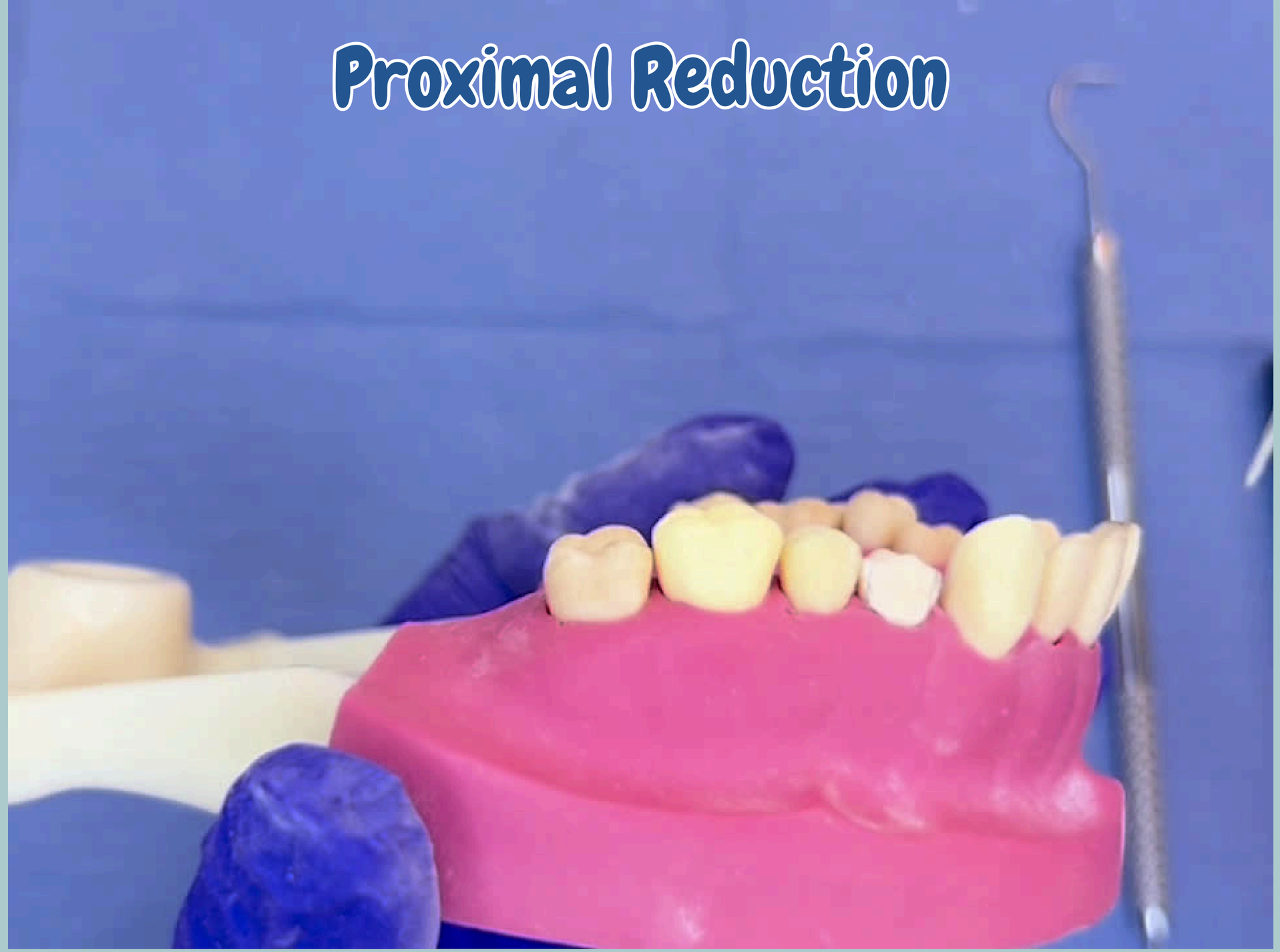
Axial reduction



Axial reduction



Proximal Reduction



The Veneered Buccal Surface



Finishing Touches

Your’e almost done! Your prep should now look something like this:



Notice how the buccal finish line is the thickest then gradually becomes thinner as we move proximally then the least thickness palatally.

The aim is that the finish line should be flushing/blended together

Finishing Touches



If you made it this far, well done! You are just a step away from being a crown prep master!

Functinal cusp bevel vs Secondary Plane

- FCB is done by lightly **bevelling/removing** the **cusp tips** of the **functional** cusps.
- Secondary plane is done on the **occlusal third** of the **buccal surface** at a 45 degree angle.

Sounds confusing we know. Just watch the upcoming videos.

Functional Cusp Bevel



Secondary Plane



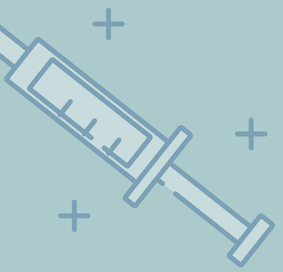
Finishing your Prep



We promise you after this step, your'e really done!

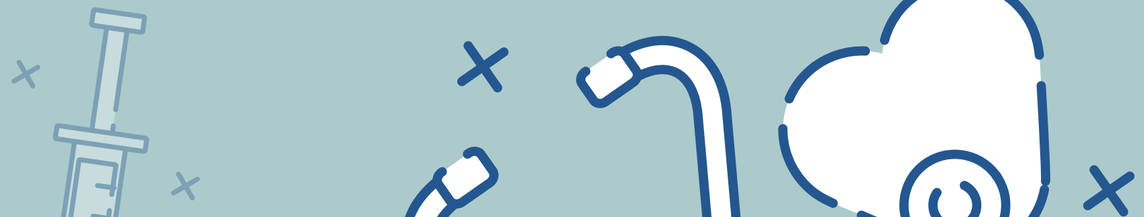


Let's talk about finishing your prep.
This step is often skipped by students thinking that it does not matter.
Let us tell you why it **does** matter:



- Blend the palatal finish line with the buccal finish line smoothly.
- Removing any sharp /undermined enamel.
- Removing irregularities provides better retention of the crown.
- Improve marginal fit of the crown.
- Reduce stress concentration.

<u>Conclusion:</u> Do **NOT** skip finishing if you want that A+ Prep!





Finishing your Prep



How to check your work

Check the AMOUNT of reduction done using index and probe	check that occlusal anatomy is preserved	check that there is no roughness or any sharp angles
Check that finish line is continous, all on the same level and has no irregularities	check that your taper is correct -no undercuts/over convergence	
Check that the thickness of buccal finish line is VISIBLY greater than the palatal finish line.	Check that both buccal and palatal finish lines blend together smoothly	
check that there is no injury to adjacent tooth	check that you did the functional cusp bevel	
check that the contact is open properly	check that secondary plane is done properly	

How to check your work




Congrats ! You have now mastered Veneered Crown Prep!

Contact Us!

If you have any further questions, stuck in any step or want to show us your progress, feel free to contact us completely free of charge!

Our talented instructor Dr. Salma is waiting for you!

Want to book a one on one session? Just give us a DM!
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https://www.tiktok.com/@dentahelp?_r=1&_t=ZS-97oeLMuArIr

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Thank
You